

# System Overview, Conventions & Hyperparameter Atlas

What the Vol-Fitter does, the pipeline, and the master index of every knob  
Vol-Fitter Technical Notes, No. 00

Vol-Fitter

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## Abstract

This is the map of the territory. The Vol-Fitter is an implied-volatility surface fitter in the spirit of a professional vola-fitter, with one differentiating capability: it *extrapolates* sparse smile observations to a full universe of smiles, across expiries *and* assets, by propagating signal through a graph whose nodes are smiles (underlying,  $T$ ). This note states the market and normalization conventions shared by the whole series, walks the end-to-end compute pipeline (fetch  $\rightarrow$  de-Americanize  $\rightarrow$  forwards  $\rightarrow$  fit  $\rightarrow$  views  $\rightarrow$  graph), catalogues what the application can do, and ends with a *master hyperparameter atlas* cross-linked to the topic notes that derive each knob. It is the front door: read it once, then any other note in the series stands alone.

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# 1 What the application is

At its core the Vol-Fitter solves one problem repeatedly — *given noisy, sparse option quotes for one expiry of one underlying, produce a smooth, arbitrage-free implied-volatility smile* — and then layers three ambitions on top of it:

1. **A choice of models, all arbitrage-aware.** Four smile representations share one calibration spine: the *LQD* log-quantile-density model (Note 01), *SVI* / *SVI-JW* (Note 02), the *Multi-Core SIV* sigmoid family (Note 03), and a full *piecewise-affine local-volatility* surface (Note 04). The first three are slice parametrizations that are butterfly-free by construction or by penalty; the fourth is a genuine local-volatility grid solving the Dupire PDE.
2. **A professional desk workflow.** Real bid/ask handling, var-swap targets, density-corrected weighting, event time dilation, dividend and forward treatment, spot-vol dynamics, and a Bayesian prior-persistence machine that decides, quote by quote, whether to trust the market or the saved prior.
3. **Graph smile-extrapolation.** The headline differentiator (Note 14): a transported prior plus the lit-node calibration innovation are propagated through an optimal-transport-regularized Bayesian graph to reconstruct the *dark* nodes nobody quoted.

## Heuristic

The unifying idea is that a volatility surface is not a curve to be drawn but a *family of probability laws* to be inferred — one risk-neutral law per (underlying,  $T$ ) node — subject to no-arbitrage couplings (butterfly within a slice, calendar across expiries) and to a Bayesian prior that ties the whole universe together. Every design choice in the app follows from taking that sentence literally.

Figure 1 shows the two structural guarantees on a four-expiry synthetic surface fitted by the production LQD calibrator: the implied densities are all non-negative (no butterfly arbitrage) and the ATM total-variance term structure is non-decreasing (no calendar arbitrage).

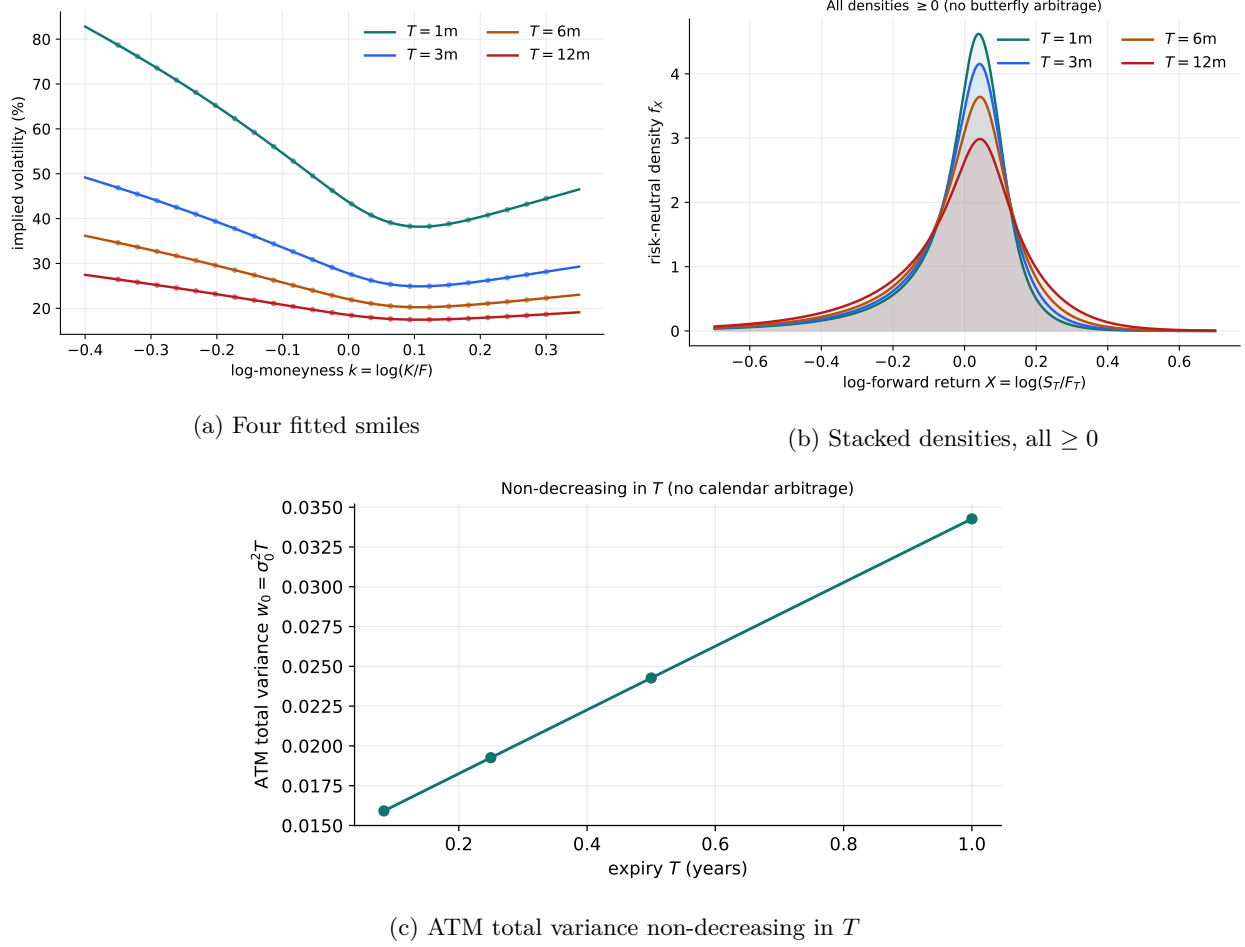


Figure 1: The two no-arbitrage guarantees, produced end to end by the production fitter on a synthetic but calendar-monotone surface.

## 2 Market conventions and normalization

These conventions hold across the entire series; individual notes assume them without restating.

**Definition 1** (Forward measure and normalization). For expiry  $T$  with forward price  $F_T$ , work under the  $T$ -forward measure and in forward-normalized, undiscounted units. The log-moneyness is  $k = \log(K/F_T)$ , the log-forward return is  $X_T = \log(S_T/F_T)$ , and normalized call/put prices are  $C_T(k) = \mathbb{E}^T[(e^{X_T} - e^k)^+]$  and  $P_T(k) = \mathbb{E}^T[(e^k - e^{X_T})^+]$ , with the martingale condition  $\mathbb{E}^T[e^{X_T}] = 1$ .

**Definition 2** (Total variance and the Black map). Total implied variance is  $w(k, T) = \sigma_{BS}^2(k, T) T$ . The normalized Black call is  $B(k, w) = \Phi(d_+) - e^k \Phi(d_-)$  with  $d_{\pm} = -k/\sqrt{w} \pm \sqrt{w}/2$ . Implied total variance is the unique  $w$  solving  $B(k, w) = C_T(k)$ ; the fitter inverts it by a vectorized scheme rather than a scalar root-find per strike.

This one primitive underlies every model in the series; in normalized form it is a handful of lines (matches production to  $10^{-12}$ ):

Listing 1: The normalized Black call  $B(k, w)$  — the shared pricing primitive (distilled from `core/black.py`).

```

1  def black_call(k, w):                                     # k = log(K/F), w = total
    variance sigma^2 T
2      sq = np.sqrt(w)
3      dp = -k/sq + 0.5*sq                                  # d_+; d_- = d_+ - sqrt(w)
4      price = norm_cdf(dp) - np.exp(k)*norm_cdf(dp - sq)
5      return np.where(w > 1e-12, price, np.maximum(1 - np.exp(k), 0.0))
    # intrinsic at w->0

```

Three conventions deserve emphasis because they recur:

- **Two clocks.** Calendar time  $t$  (year-fraction to expiry) and *variance* time  $\tau$  (calendar time dilated by scheduled events) are kept separate. Fits live on  $\tau$ ; implied volatility is reported as  $\sqrt{w/\tau}$ , so adding an event *lowers* reported IV while preserving price (Note 11).
- **Forward-normalized everything.** Prices, strikes and densities are quoted relative to  $F_T$ , so a single smile object is spot-independent until a dynamics rule (Note 12) transports it under a spot move.
- **Vega-normalized residuals.** Calibration losses are price residuals divided by Black vega, which makes the objective behave like a volatility error while keeping every iterate a valid price (Notes 01, 04, 07).

### 3 The compute pipeline

Every smile the app shows is the output of the pipeline in fig. 2. The stages are deliberately decoupled and independently cached so that, e.g., editing a fit penalty never re-pulls data or re-runs de-Americanization.

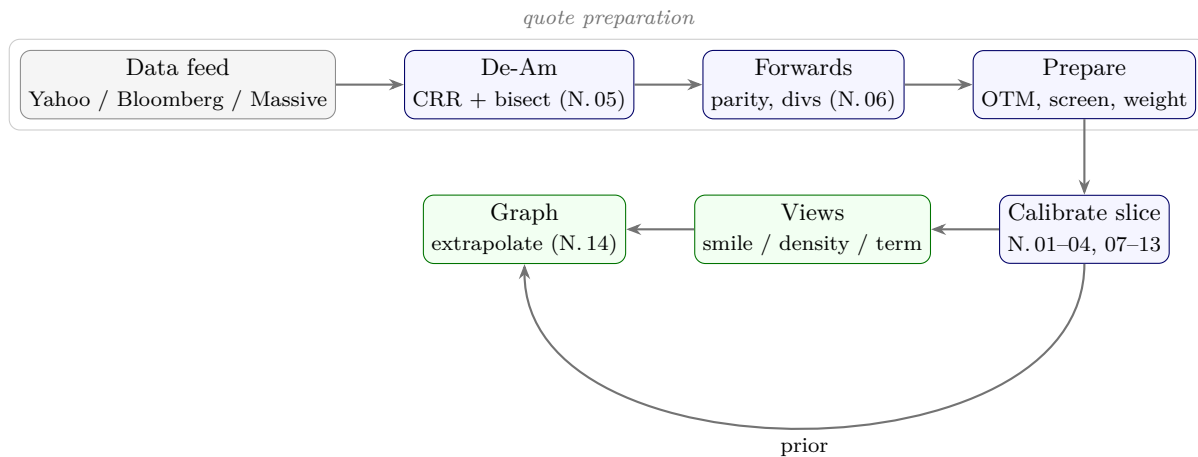


Figure 2: The end-to-end compute spine. Quote preparation (de-Americanize, forwards/dividends, OTM selection, static screen, weighting) feeds the per-slice calibration; the calibrated lit nodes drive both the viewer and the graph extrapolator, which reconstructs the dark nodes.

## Stage by stage.

1. **Data feed.** A provider registry exposes Yahoo (US options), Bloomberg (xbbg; US + non-US indices/equities), and Massive (Polygon/OPRA), plus a synthetic source; an as-of selector serves Live / previous-close / EOD / captured-replay. US-options-only providers and the non-US Bloomberg path are handled transparently.
2. **De-Americanization** (Note 05). Listed equity options are American; each quote is converted to its European-equivalent implied volatility by inverting a Cox–Ross–Rubinstein tree (bisection), with a Numba kernel for wide chains and a content-digest cache so fit-tuning never re-runs it. A wing-only, band-constrained convexity repair then removes the butterfly arbitrage the per-strike inversion can leave in the sparse wings, holding the ATM core fixed.
3. **Forwards and dividends** (Note 06). Put–call parity yields the implied forward and discount (clamped against feed noise); discrete dividends and American de-bias are handled here.
4. **Prepare.** Out-of-the-money selection, a static no-arbitrage/wing screen, and the choice of observation weights (Note 07).
5. **Calibrate slice.** The chosen model is fitted to the prepared quotes under the active objective: mid / bid–ask band / haircut (Note 07), optional var-swap target (Note 08), calendar coupling (Note 10), event clock (Note 11), and Bayesian prior persistence (Note 13).
6. **Views.** Every smile-derived view — smile, stacked densities, stacked total variance, quantile density, term structure, local-vol surface, SSR scenario — follows the chosen model and is downsampled for the UI.
7. **Graph** (Note 14). The transported prior plus the lit-node innovation propagate through the OT-regularized Bayesian graph to reconstruct dark-node smiles with credible bands.

## 4 The smile graph

The object that makes the app more than a slice fitter is the *smile graph* (fig. 3): nodes are smiles (underlying,  $T$ ); edges encode that one node informs another. Calendar edges link expiries of one name; cross-asset edges link an index or sector ETF to its single names. A user marks nodes *lit* (quoted, calibrated) or *dark* (to be reconstructed). The graph carries the three ATM handles ( $\sigma_0, s_0, \kappa_0$ ) of each node as a Gaussian signal and propagates the lit innovation to the dark nodes; Note 14 develops the optimal-transport-regularized Bayesian solver behind it, and the leave-one-out backtest that measures its skill.

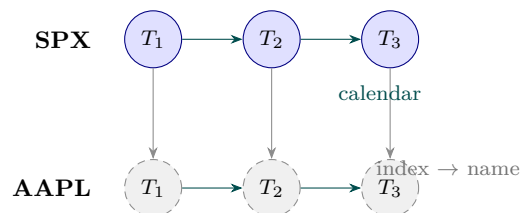


Figure 3: A fragment of the smile graph: lit index nodes (solid) inform dark single-name nodes (dashed) across both calendar and cross-asset edges. The graph posterior reconstructs the dark smiles from the lit innovation.

## 5 Capability catalogue

Table 1 indexes what the app can do against the note that documents the mathematics. It doubles as a reading order.

Table 1: Capability  $\rightarrow$  note index.

| Capability            | Summary                                                                | Note |
|-----------------------|------------------------------------------------------------------------|------|
| LQD smile model       | Arbitrage-free-by-construction log-quantile density; analytic Jacobian | 01   |
| SVI / SVI-JW          | Raw and jump-wing SVI with no-arb penalty and Lee cap                  | 02   |
| Multi-Core SIV        | Sigmoid base + zero-wing hats for WW/event smiles                      | 03   |
| Local volatility      | Piecewise-affine Dupire surface; GN/Numba calibration                  | 04   |
| De-Americanization    | CRR tree + bisection, Numba kernel, cache digest, wing convex repair   | 05   |
| Forwards / dividends  | Parity forward + discount clamp, discrete divs                         | 06   |
| Weighting & bid-ask   | Equal / time-value-density weights; band/haircut fit                   | 07   |
| Variance-swap targets | Replication, LQD closed form, source-PDE                               | 08   |
| Wings & Lee bounds    | Tail control unified across models                                     | 09   |
| Calendar arbitrage    | Convex-order soft hinge, model-agnostic                                | 10   |
| Event variance clock  | Dual calendar/variance time, day-weighted events                       | 11   |
| Spot-vol dynamics     | Sticky strike/moneyness/local-vol, SSR transport                       | 12   |
| Prior persistence     | 7-mode Bayesian menu, precision gate, two-pass                         | 13   |
| Graph extrapolation   | Transported prior + directed increments + back-test                    | 14   |

**Viewer and data capabilities (engineering, not derived here).** The workspace is organized into Parametric, LocalVol, Forwards, Options, Graph and Universe tabs: charts of prior/current fit vs. quote bands in normalized or fixed strike, quantile and density functions, term structure and event-dilated calendar in vol and variance, var-swap levels, interactive quote select/erase/amend, universe selection, lit/dark node maps, and the graph editor. Data plumbing (providers, as-of replay, dividend editor, persistence of named universes and fit history) is described in the project README rather than in this mathematical series.

## 6 How the notes are organized

Each note is standalone and follows the same skeleton: assumptions and heuristics first, then full derivations in a clean-math body, then a worked example with figures generated by the production code, and finally two appendices — **A, a hyperparameter atlas** (every knob, surfaced or hidden, with its default) and **B, performance** (the numerical optimizations and measured timings). Notes cross-reference each other (this note’s table 1 is the index) but never depend on one another’s

internals. Numbers and plots are *measured*, not copied: where a fresh run differs from an older design note, the fresh run wins.

## A Master hyperparameter atlas

The complete index of tunable parameters across the app, surfaced (`FitSettings` / `OptionsSettings`, user-facing) and hidden (internal constants). Each row points to the note that derives its role; defaults are the shipped values. This is a directory, not a derivation — consult the referenced note for meaning.

Table 2: Surfaced parameters.

| Parameter                                      | Default          | Role                                        | Note   |
|------------------------------------------------|------------------|---------------------------------------------|--------|
| <code>nOrder</code>                            | 6                | LQD Legendre order                          | 01     |
| <code>regLambda</code> / <code>regPower</code> | $10^{-6}$ / 1    | LQD high-order ridge                        | 01     |
| <code>barrierCenter</code>                     | / 0.90 / 50      | LQD $A_R$ barrier                           | 01     |
| <code>barrierScale</code>                      |                  |                                             |        |
| <code>nCores</code>                            | 2                | Multi-Core SIV hat count (hard-capped at 2) | 03     |
| <code>sigmoidRidge</code>                      | $10^{-2}$        | SIV hat-amplitude ridge                     | 03     |
| <code>sivWingPenaltyPct</code>                 | 100              | SIV put-wing Durrleman penalty (0 = off)    | 03     |
| <code>sviPenaltyWeight</code>                  | $10^3$           | SVI no-arbitrage penalty                    | 02     |
| <code>leeSlopeMax</code>                       | 2.0              | SVI Lee wing-slope cap                      | 02, 09 |
| <code>gridXNodes</code>                        | 12               | LV strike-grid vertices                     | 04     |
| <code>gridXMinPerExpiry</code>                 | 8                | LV short-dated coverage floor               | 04     |
| <code>gridTNodes</code>                        | 10               | LV maturity-grid vertices                   | 04     |
| <code>gridRegLambda</code>                     | / $10^{-2}$ / 1  | LV roughness penalty / ratio                | 04     |
| <code>gridRegRho</code>                        |                  |                                             |        |
| <code>convexWingWeight</code>                  | $10^3$           | LV wing-convexity penalty                   | 04     |
| <code>frontTieWeight</code>                    | $10^{-2}$        | LV front-end tie-to-ATM                     | 04     |
| <code>lvVolCapMult</code>                      | 3.0              | LV local-vol cap multiple                   | 04     |
| <code>weightScheme</code>                      | equal            | equal / tv-density weights                  | 07     |
| <code>fitMode</code>                           | mid              | mid / bidask / haircut                      | 07     |
| <code>haircut</code>                           | 0.005            | band haircut (vol)                          | 07     |
| <code>midAnchorWeight</code>                   | 0.05             | band mid anchor                             | 07     |
| <code>varSwapWeightPct</code>                  | 10%              | var-swap budget                             | 08     |
| <code>calendarWeight</code>                    | $10^6$           | calendar soft-constraint penalty            | 10     |
| <code>ssr</code>                               | 2.0              | spot-vol SSR / regime                       | 12     |
| <code>priorPersistenceMode</code>              | hybrid           | prior mode selector                         | 13     |
| <code>priorOperatorSet</code>                  | ATM/RR25/BF25/VR | operators to persist                        | 13     |
| <code>priorOperatorStrengthPct</code>          | 50%              | operator prior budget                       | 13     |
| <code>priorOperatorRequiredPrecision</code>    | 1.0              | operator activation gate                    | 13     |
| <code>priorOperatorGapExponent</code>          | 1.0              | gate exponent $\gamma$                      | 13     |
| <code>priorOperatorBandwidth</code>            | 0.06             | operator leg KDE bandwidth                  | 13     |

| Parameter                  | Default          | Role                       | Note |
|----------------------------|------------------|----------------------------|------|
| priorFactorSet             | ATM/skew/curv/VS | stable factors to persist  | 13   |
| priorFactorStrengthPct     | 50%              | factor prior budget        | 13   |
| priorTailAnchorStrengthPct | 20%              | hybrid tail-anchor budget  | 13   |
| priorAnchorWeightPct       | 50%              | strike-gap prior budget    | 13   |
| graphKappaScale $\kappa$   | 1.0              | graph temporal precision   | 14   |
| graphEtaScale $\eta$       | 1.0              | graph directed smoothness  | 14   |
| graphLambdaScale $\lambda$ | 0.0              | graph OT regularization    | 14   |
| graphNu $\nu$              | 0.1              | graph UOT source allowance | 14   |
| events                     | —                | scheduled event clock      | 11   |

Table 3: Hidden internal constants (selected).

| Constant               | Default       | Role                            | Note |
|------------------------|---------------|---------------------------------|------|
| Z_MAX / N_POINTS       | 40 / 8001     | LQD quadrature grid             | 01   |
| OPT_N_POINTS           | 2001          | LQD optimization grid           | 01   |
| EPS_AR                 | $10^{-6}$     | LQD $A_R < 1$ buffer            | 01   |
| DEFAULT_N_K            | 1201          | LV PDE strike nodes             | 04   |
| DEFAULT_DT_MAX         | 1/400         | LV PDE max time step            | 04   |
| N_RANNACHER            | 4             | LV Rannacher start steps        | 04   |
| VAR_FLOOR              | $10^{-6}$     | Dupire local-variance floor     | 04   |
| DEFAULT_STEPS          | 501           | CRR scalar tree depth           | 05   |
| DEFAULT_BATCH_STEPS    | 192           | CRR batch tree depth            | 05   |
| BATCH_BISECTIONS       | 24            | de-Am bisection sweeps          | 05   |
| SIGMA_LO / SIGMA_HI    | $10^{-4}$ / 4 | implied-vol bracket             | 05   |
| VS_HALF_WIDTH          | / 6 / 801     | var-swap replication grid       | 08   |
| VS_POINTS              |               |                                 |      |
| CAL_STRIDE             | 25            | calendar constraint stride      | 10   |
| DAYS_PER_YEAR          | 365           | calendar→variance clock         | 11   |
| SPREAD_HALF            | 0.05          | precision spread half-width     | 13   |
| OBS_FRESHNESS_HALFLIFE | 3 d           | observation freshness decay     | 13   |
| TRANSPORT_SCALE        | 0.10          | prior transport precision decay | 13   |
| HANDLE_CONFIDENCE      | [1, 1, 0.01]  | per-handle graph confidence     | 14   |
| SOURCE_BASE            | tiered        | prior provenance precision      | 14   |

**Reading the atlas.** A surfaced parameter is a *policy* choice — how strong a penalty, which prior mode, how to weight quotes — safe to tune. A hidden constant is tied to an algorithm’s correctness (the  $A_R$  buffer, the PDE floor, the tree depth) and is exposed only in the relevant note’s Appendix A, where the consequences of changing it are spelled out. Together they are the full control surface of the fitter.